

#BLACKBOX CTF — Solver Write-up

Challenge

****Name:**** The Broken License

****Category:**** Reverse Engineering

****Flag:**** `TECH{reverse\_the\_blackbox}`

1. Reconnaissance

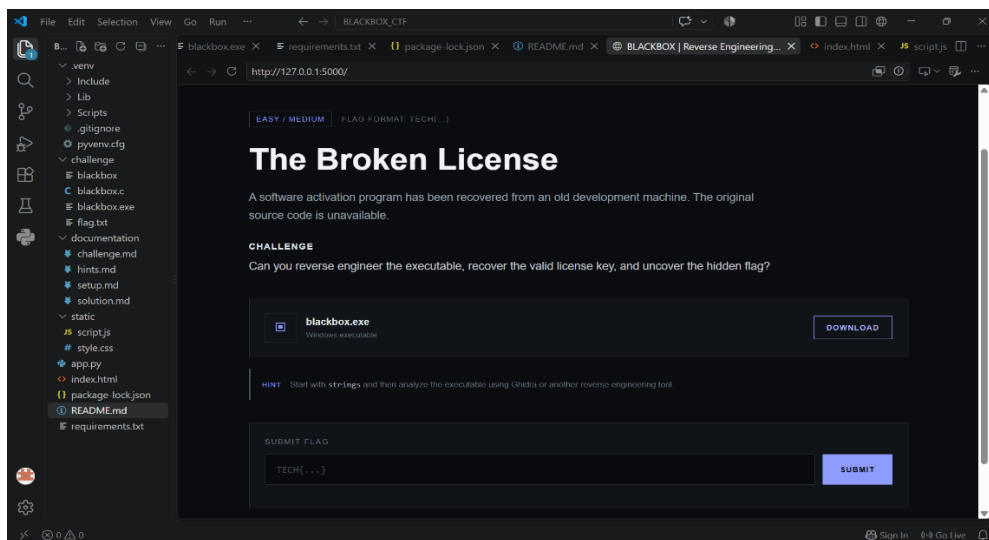
I was provided with a Windows executable named `blackbox.exe`.

I first executed it using:

```
` `` powershell
```

.\blackbox.exehe program asked for a license key.

Screenshot 1:



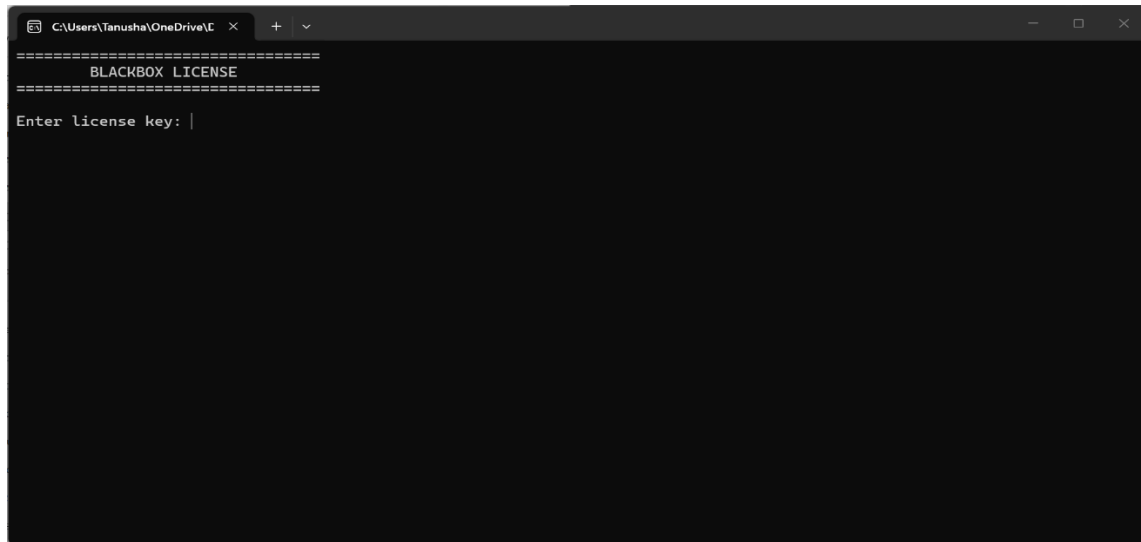
2. Testing

I entered an incorrect value such as:

TEST

The program rejected the input, which indicated that a license validation mechanism was present.

Screenshot 2:



3. Reverse Engineering

I imported blackbox.exe into **Ghidra** and performed static analysis.

I first inspected the strings and searched for messages such as:

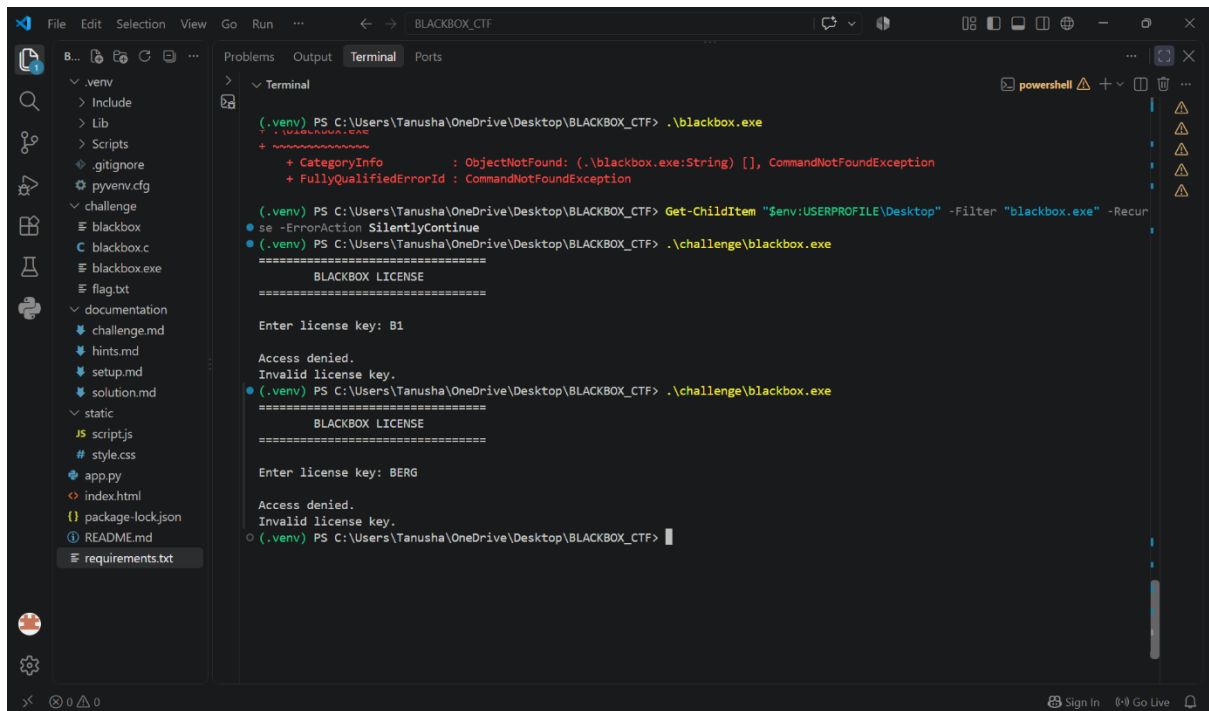
Enter license key

Access granted

Access denied

I then followed the references to locate the license-validation logic.

Screenshot 3:



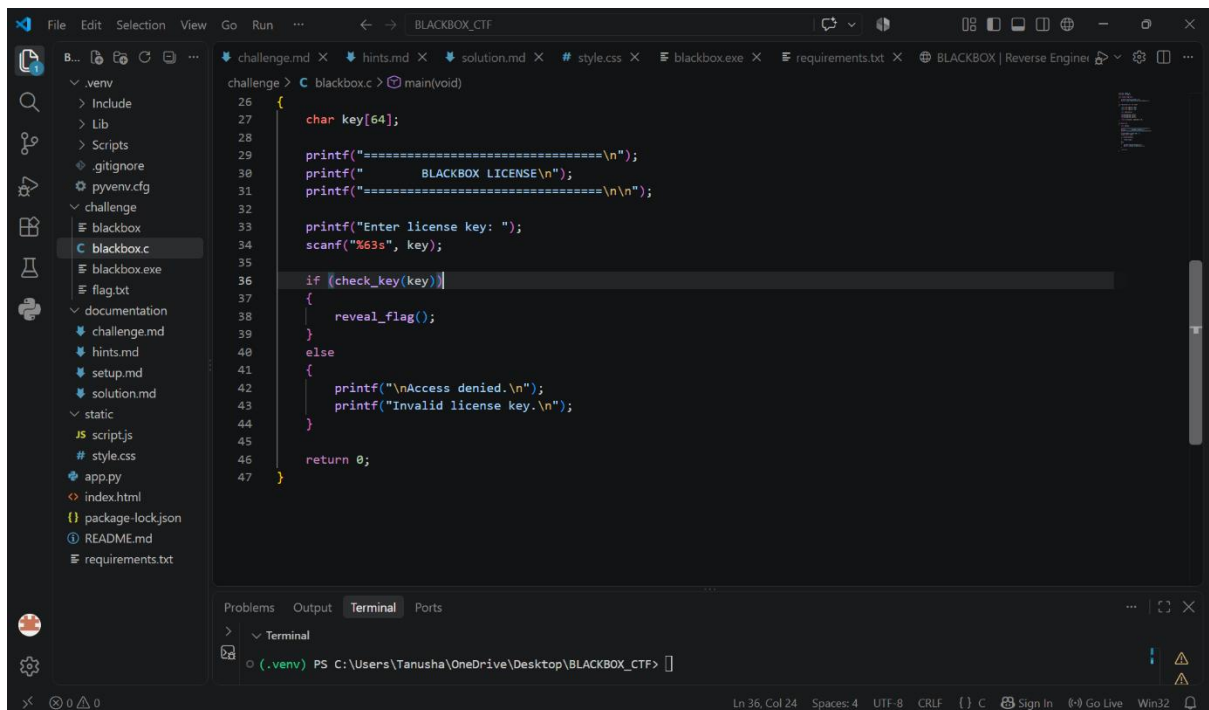
```
(.venv) PS C:\Users\tanusha\OneDrive\Desktop\BLACKBOX_CTF> .\blackbox.exe
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (.\blackbox.exe:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

(.venv) PS C:\Users\tanusha\OneDrive\Desktop\BLACKBOX_CTF> Get-ChildItem "$env:USERPROFILE\Desktop" -Filter "blackbox.exe" -Recur
se -ErrorAction SilentlyContinue
(.venv) PS C:\Users\tanusha\OneDrive\Desktop\BLACKBOX_CTF> .\challenge\blackbox.exe
=====
BLACKBOX LICENSE
=====
Enter license key: B1

Access denied.
Invalid license key.
(.venv) PS C:\Users\tanusha\OneDrive\Desktop\BLACKBOX_CTF> .\challenge\blackbox.exe
=====
BLACKBOX LICENSE
=====
Enter license key: BERG

Access denied.
Invalid license key.
(.venv) PS C:\Users\tanusha\OneDrive\Desktop\BLACKBOX_CTF>
```

Screenshot 4:



```
challenge > C:\blackbox.c> main(void)
26
27     char key[64];
28
29     printf("=====\n");
30     printf("BLACKBOX LICENSE\n");
31     printf("=====\n");
32
33     printf("Enter license key: ");
34     scanf("%63s", key);
35
36     if (check_key(key))
37     {
38         reveal_flag();
39     }
40     else
41     {
42         printf("\\nAccess denied.\\n");
43         printf("Invalid license key.\\n");
44     }
45
46     return 0;
47 }
```

4. Recovering the Key

The validation logic contained three separate key fragments:

B1

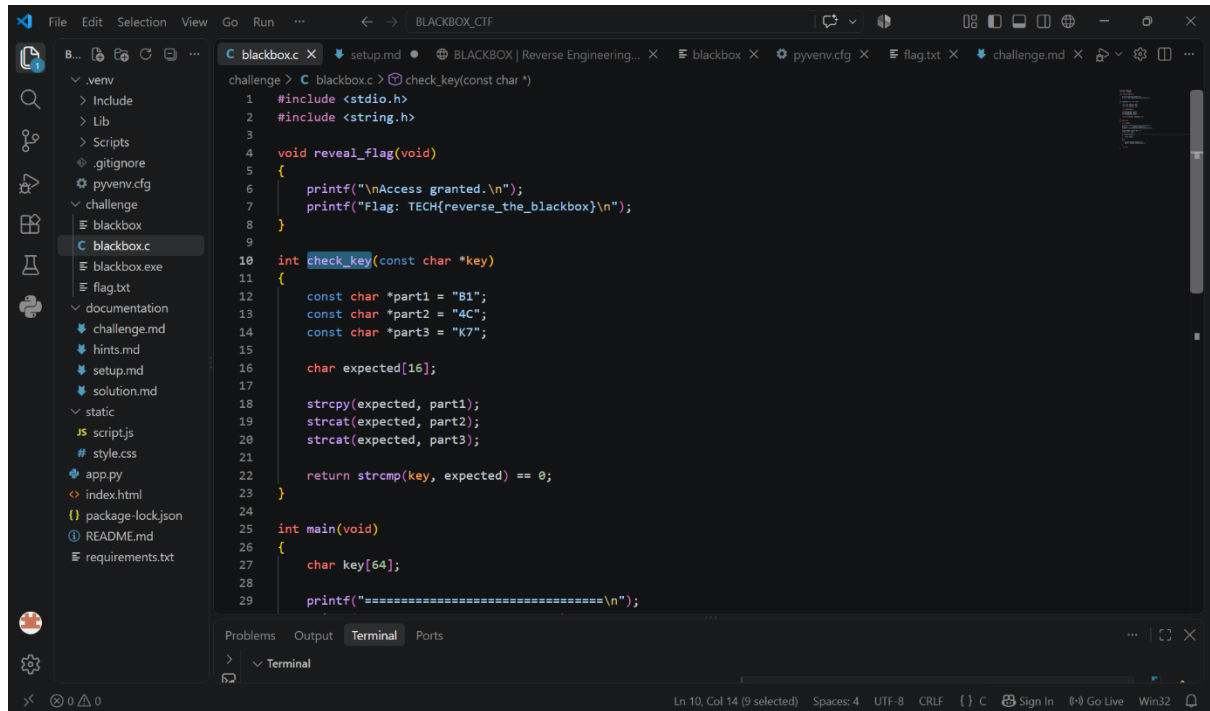
4C

K7

Combining them in the required order gave:

B14CK7

Screenshot 5:



```
challenge > C blackbox.c > C check_key(const char *)
1  #include <stdio.h>
2  #include <string.h>
3
4  void reveal_flag(void)
5  {
6      printf("\nAccess granted.\n");
7      printf("Flag: TECH{reverse_the_blackbox}\n");
8  }
9
10 int check_key(const char *key)
11 {
12     const char *part1 = "B1";
13     const char *part2 = "4C";
14     const char *part3 = "K7";
15
16     char expected[16];
17
18     strcpy(expected, part1);
19     strcat(expected, part2);
20     strcat(expected, part3);
21
22     return strcmp(key, expected) == 0;
23 }
24
25 int main(void)
26 {
27     char key[64];
28
29     printf("=====\n");
```

5. Verification

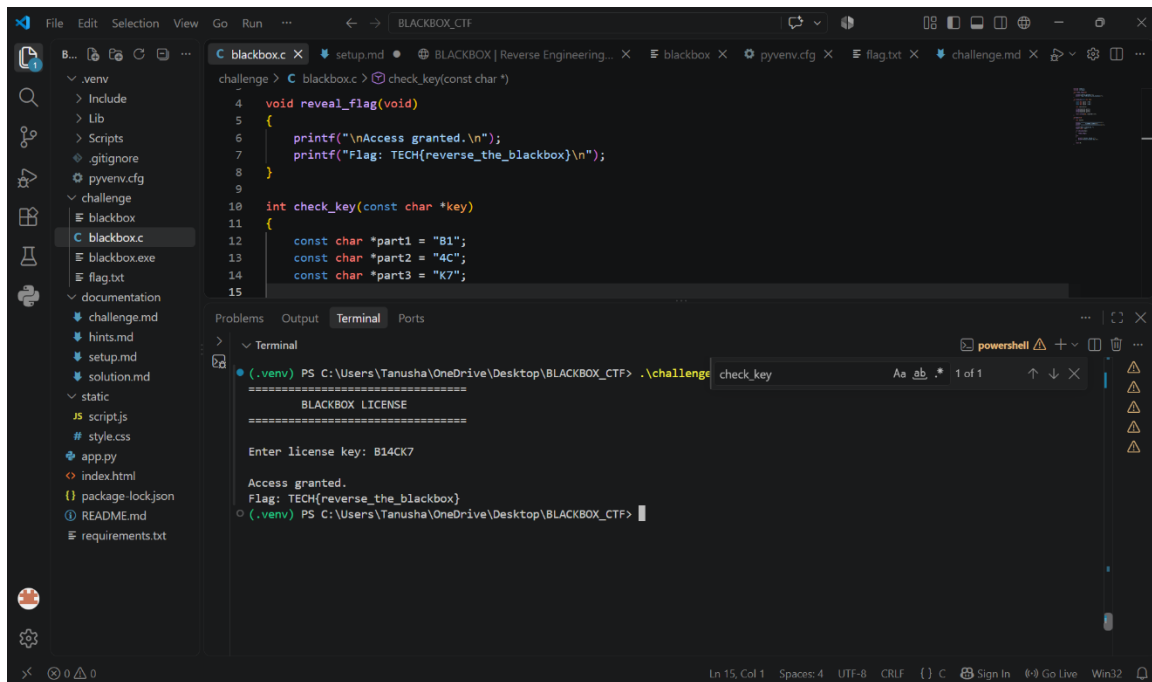
I ran the executable again and entered:

B14CK7

The program accepted the key and revealed:

TECH{reverse_the_blackbox}

Screenshot 6:



The screenshot shows a VS Code editor with a project named 'BLACKBOX_CTF'. The file explorer on the left shows a directory structure with files like 'blackbox.c', 'blackbox.exe', 'flag.txt', and 'challenge.md'. The main editor window displays the source code of 'blackbox.c', which includes a 'reveal_flag' function and a 'check_key' function. The terminal at the bottom shows the execution of the program, which prompts for a license key and displays the flag 'TECH{reverse_the_blackbox}'.

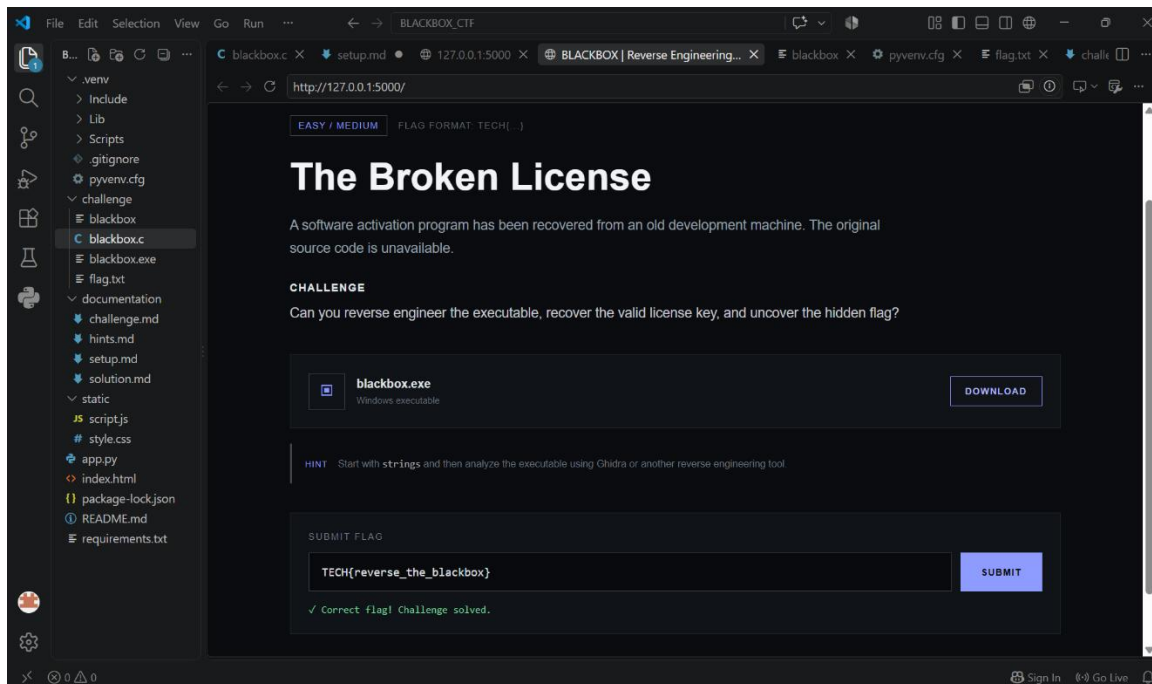
```
challenge > C:\blackbox> .\check_key(const char *)
4 void reveal_flag(void)
5 {
6     printf("\nAccess granted.\n");
7     printf("Flag: TECH{reverse_the_blackbox}\n");
8 }
9
10 int check_key(const char *key)
11 {
12     const char *part1 = "B1";
13     const char *part2 = "4C";
14     const char *part3 = "K7";
15 }
```

```
(.venv) PS C:\Users\tanusha\OneDrive\Desktop\BLACKBOX_CTF> .\challenge check_key
=====
BLACKBOX LICENSE
=====
Enter license key: B14CK7
Access granted.
Flag: TECH{reverse_the_blackbox}
(.venv) PS C:\Users\tanusha\OneDrive\Desktop\BLACKBOX_CTF>
```

6. Flag Submission

I entered the recovered flag into the CTF web interface and received a successful submission message.

Screenshot 7:



Tools Used

- PowerShell
- Ghidra
- GCC/MinGW

Methodology

Run binary

→ Test input

→ Analyze with Ghidra

→ Find validation logic

→ Recover key

→ Verify key

→ Obtain flag

→ Submit flag

Conclusion

The challenge demonstrated how static analysis can be used to understand an executable's validation logic and recover information hidden inside a compiled program.